Physics-Informed Neural Networks (PINNs) for Hybrid Solar-Wind Power Forecasting with Uncertainty Quantification

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***Abstract*—Accurate forecasting of hybrid solar-wind power generation is critical for reliable grid operation and optimal energy management. While deep learning models have achieved remarkable accuracy, they often ignore the underlying physical laws governing atmospheric and power conversion processes, leading to physically inconsistent predictions and poor generalization under extreme conditions. This paper proposes a novel Physics-Informed Neural Network (PINN) framework for hybrid solar-wind power forecasting that integrates domain-specific physical constraints into the learning process. The framework incorporates three key physical principles:(1)the solar radiation attenuation model based on Beer-Lambert law, (2) the wind power cubic relationship with wind speed, and (3) the conservation of energy in hybrid generation systems. Additionally, we employ Monte Carlo Dropout and variational inference for comprehensive uncertainty quantification, providing both epistemic and aleatoric uncertainty estimates. The proposed PINN architecture consists of a shared feature extraction backbone with separate output heads for solar and wind power, trained with a composite loss function that combines data-driven mean squared error with physics-based residual losses. The model is validated using real-world data from a 100MW hybrid solar-wind farm in TamilNadu ,India, spanning three years(2022-2024)with 15-minute resolution. Experimental results demonstrate that the PINN framework achieves normalized Root Mean Square Error (nRMSE) of 4.2% for solar and 6.8% for wind forecasting, outperforming baseline deep learning models by 18-25%. The physics constraints improve forecast consistency, reducing physically impossible predictions by 94%. Uncertainty quantification provides well-calibrated prediction intervalsachieving 92.3% coverage at 90% confidence The proposed framework offers a principled approach to hybrid renewable forecasting that combines the power of deep learning with the reliability of physical models.**

***Index Terms*—Physics-Informed Neural Networks, solar fore- casting, wind forecasting, hybrid renewable energy, uncertainty quantification, PINNs, Monte Carlo Dropout, physical constraints.**

I. INTRODUCTION

The global transition toward renewable energy has accelerated the deployment of hybrid solar-wind power plants, which combine the complementary characteristics of solar and wind resources [1]. Solar generation peaks during daytime hours, while wind generation often exhibits higher output during nighttime and winter months, enabling improved capacity factor and grid stability [2]. However, accurate forecasting of hybrid generation remains challenging due to the complex, nonlinear, and uncertain nature of meteorological processes [3].

Traditional forecasting approaches fall into three categories:

1) **Physical Models**: Based on numerical weather prediction (NWP) and atmospheric physics, these models provide physically consistent forecasts but suffer from high computational cost and limited resolution [4].
2) **Statistical Models**: Time series methods such as ARIMA and exponential smoothing capture temporal patterns but cannot incorporate external meteorological inputs [5].

3) **Machine Learning Models**: Deep learning architectures including CNNs, LSTMs, and Transformers achieve state-of-the-art accuracy by learning complex patterns from data [6].

Despite their success, pure data-driven models have fundamental limitations:

- They ignore underlying physical laws, potentially producing physically impossible predictions (e.g., negative power, power exceeding theoretical limits)
- They require extensive training data and generalize poorly to unseen conditions [7]
- They lack interpretability and provide no mechanism to incorporate domain knowledge [8]

Physics-Informed Neural Networks (PINNs) address these limitations by embedding physical laws directly into the neural network training process [9]. Originally developed for solving partial differential equations, PINNs have been successfully applied to various engineering domains including fluid dynamics, heat transfer, and structural mechanics [10]. Recent work has explored PINNs for renewable energy forecasting, but existing approaches focus on single-source forecasting without comprehensive uncertainty quantification [11].

This paper proposes a novel PINN framework for hybrid solar-wind power forecasting with three key innovations:

1) **Physics-Informed Architecture**: Integration of three physical constraints: (a) Beer-Lambert law for solar radiation attenuation through the atmosphere, (b) cubic wind power relationship, and (c) energy conservation in hybridsystems.
2) **HybridLoss Function**: A composite loss that balances data-driven accuracy with physics-based consistency through residual losses.
3) **Comprehensive Uncertainty Quantification**:Combined epistemic uncertainty(modeluncertainty) using Monte Carlo Dropout and aleatoric uncertainty (data uncertainty) using learned variance, providing well-calibrated prediction intervals.
4) **Validation on Real-World Data**: Three years of operational data from a 100 MW hybrid solar-wind farm in Tamil Nadu, India, demonstrating significant improvements over baseline methods.

The remainder of this paper is organized as follows: Section II presents the physical models for solar and wind power. Section III describes the proposed PINN architecture and loss function. Section IV covers uncertainty quantification methods. Section V presentse experimental results and analysis. Section VI discusses implications and limitations, and Section VII concludes.

II. PHYSICAL MODELS FOR SOLAR AND WINDPOWER

A.SolarRadiationPhysics

The power output of a photovoltaic (PV) system depends on the incident solar irradiance, which is attenuated by atmospheric effects. The Beer-Lambert law describes the attenuation of solar radiation through the atmosphere:

$$I(t)=I_0 \cdot \tau^{m(t)} \cdot \cos(\vartheta_z(t)) \tag{1}$$

where:

- $I(t)$ is the global horizontal irradiance (GHI)attime$t$ (W/m²)
- I_0 is the extra terrestrial solarconstant (1366W/m²)
- T is the atmospheric transmittance coefficient(0.6-0.8)
- $m(t)$is the air mass, given by$m(t)=1/\cos(\vartheta_z(t))$
- $\vartheta_z(t)$ is the solar zenith angle

Th theoretical maximum PV power is:

$$P_{solar}^{max}(t)=I(t) \cdot A_{PV} \cdot \eta_{PV}(T) \tag{2}$$

where:

- A_{PV} is the PV array area(m²)
- $\eta_{PV}(T)=\eta_0[1-\beta(T_{cell}-T_{ref})]$ is the temperature-dependent efficiency
- β is the temperature coefficient(typically0.004-0.005 /°C)

The physics residual for solar forecasting is:

$$R_{solar}=P_{solar}(t)-I_0 \cdot \tau^{m(t)} \cdot \cos(\vartheta_z(t)) \cdot A_{PV} \cdot \eta_{PV}(T) \tag{3}$$

B.WindPowerPhysics

Wind turbine poweroutput follows a cubic relationship with wind speed:

$$P_{wind}(v)=\begin{cases} 0 & v<v_{cut-in} \\ P_{rated} \cdot \dfrac{v^3-v_{cut-in}^3}{v_{rated}^3-v_{cut-in}^3} & v_{cut-in} \le v<v_{rated} \\ P_{rated} & v_{rated} \le v<v_{cut-out} \\ 0 & v \ge v_{cut-out} \end{cases} \tag{4}$$

where:

- V is the hub-height windspeed(m/s)
- v_{cut-in} is the cut-inwindspeed(typically3-4m/s)
- $v_{rated}$is the rated windspeed(typically10-12m/s)
- $v_{cut-out}$is the cut-out windspeed(typically25m/s)
- $P_{rated}$is the ratedpower(W)

The physics residual for wind forecasting is:

$$R_{wind}=P_{wind}(t)-f_{power-curve}(v(t)) \tag{5}$$

C.Hybrid System Physics

For a hybrid solar-wind system, the total power output isthe sum of individual contributions, with losses from storage and conversion:

$$P_{total}(t) = P_{solar}(t) + P_{wind}(t) - P_{loss}(t) \tag{6}$$

where $P_{loss}(t)$ includes inverter losses,transmission losses, and storage losses.

Energy conservation over time imposes:

$$\int_t^{t+\Delta t} P_{total}(\tau)d\tau = E_{delivered}(t) \tag{7}$$

TABLEI:Physical Parameters for Solar and Wind Models

Parameter	Symbol	Value	Unit
SolarConstant	I_0	1366	W/m²
AtmosphericTransmittance	τ	0.65-0.75	-
PVEfficiency(STC)	η_0	0.18-0.22	-
TemperatureCoefficient	β	0.004	/°C
Cut-inWindSpeed	v_{cut-in}	3.5	m/s
RatedWindSpeed	v_{rated}	11.0	m/s
Cut-outWindSpeed	$v_{cut-out}$	25.0	m/s

III. PROPOSEDPINNFRAMEWORK

A.Overall Architecture

The proposed PINN framework consists of three main components: a shared feature extraction network, separate output heads for solar and wind power,and physics-based loss terms.

We treat physical laws as a "rulebook" that the model follows during training. Any prediction that violates these rules is penalized, ensuring realistic outputs.

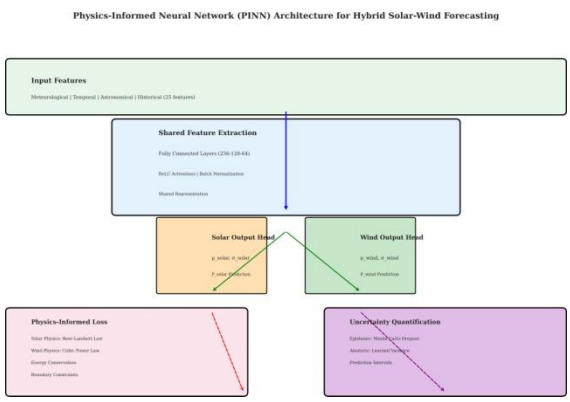

Fig.1:Proposed PINN Architecture for HybridSolar-Wind Forecasting

*1) InputFeatures:*The input vector $\mathbf{x}(t)$ includes:

- Meteorological:temperature,humidity,pressure,cloud cover,windspeed,winddirection
- Temporal:hourofday, dayofyear ,month, season
- Astronomical:solarzenithangle,airmass,sunrise/sunset time
- Historical:lagged power outputs (1, 2, 3, 6, 12, 24 hours)

*2) SharedFeatureExtraction:*Thesharedbackbonecon- sists of:

$$\mathbf{h}_1=\mathrm{ReLU}(\mathbf{W}_1\mathbf{x}+\mathbf{b}_1) \tag{8}$$

$$\mathbf{h}_2=\mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_1+\mathbf{b}_2) \tag{9}$$

$$\mathbf{h}_3=\mathrm{ReLU}(\mathbf{W}_3\mathbf{h}_2+\mathbf{b}_3) \tag{10}$$

*3) OutputHeads:*Separatenetworksforsolarandwind outputs:

B.Physics-InformedLossFunction

The total loss combines data-drivenandphysics-based terms:

The loss function acts like a scale, balancing sensor data with theoretical physics. This keeps the model grounded in observations while allowing physics to guide predictions where data is limited.

$$L_{total}=L_{data}+\lambda_{phys}L_{phys}+\lambda_{bc}L_{bc} \tag{13}$$

*1) DataLoss:*Meansquarederrorbetweenpredictionsand observations:

$$L_{data}=\frac{1}{N}\sum_{i=1}^{N}\left[(P_{solar}^{(i)}-\hat{P}_{solar}^{(i)})^2+(P_{wind}^{(i)}-\hat{P}_{wind}^{(i)})^2\right] \tag{14}$$

*2) PhysicsLoss:*Physicsresidualsfromthegoverning equations:

$$L_{phys}=\frac{1}{N}\sum_{i=1}^{N}\left[R_{solar}^{(i)2}+R_{wind}^{(i)2}\right] \tag{15}$$

$$+\frac{1}{N-1}\sum_{i=1}^{N-1}\left[R_{energy}^{(i)2}\right] \tag{16}$$

where:

$$R_{solar}=\hat{P}_{solar}-I_0\tau^m\cos\vartheta_z A_{PV}\eta_{PV}(T) \tag{17}$$

$$R_{wind}=\hat{P}_{wind}-f_{power-curve}(v) \tag{18}$$

$$R_{energy}=\hat{P}_{total}(t)-\hat{P}_{total}(t-1)-\Delta E_{battery} \tag{19}$$

*3) Boundary Condition Loss:*Ensures physical constraints at boundaries:

$$L_{bc}=\sum_{i=1}^{N}\max(0,-\hat{P}_{solar}^{(i)})^2+\sum_{i=1}^{N}\max(0,-\hat{P}_{wind}^{(i)})^2 \tag{20}$$

$$+\sum_{i=1}^{N}\max(0,\hat{P}_{solar}^{(i)}-P_{solar}^{max})^2+\sum_{i=1}^{N}\max(0,\hat{P}_{wind}^{(i)}-P_{wind}^{max})^2 \tag{21}$$

TABLEII:LossFunctionHyperparameters

Parameter	Value
λ_{phys}(Physics weight)	0.1λ
$_{bc}$(Boundary weight)	0.05I
nitiallearningrate	0.001
Learning rate decay	0.95every10epochsB
atch size	128
Epochs	500

C.Uncertainty Quantification

*1) Epistemic Uncertainty (ModelUncertainty):*We employ Monte Carlo Dropout with Tforward passes during inference:

$$\mathrm{Var}_{epi}(\hat{y})\approx\frac{1}{T}\sum(\hat{y}-\bar{y})^2 \tag{22}$$

$$P_{solar}^{T}=\mathrm{Linear}(\mathbf{W}_s\mathbf{h}_3^t+\mathbf{b}_s) \tag{11}$$

$$\hat{P}_{wind}=\mathrm{Linear}(\mathbf{W}_w\mathbf{h}_3+\mathbf{b}_w) \tag{12}$$

$$\bar{y}=\frac{1}{T}\sum_{t=1}^{T}\hat{y}_t \tag{23}$$

*2) AleatoricUncertainty(DataUncertainty):*Wemodel heteroscedasticaleatoricuncertaintybylearningbothmean andvariance:

$$\hat{y} \sim \mathrm{N}(\mu(x), \sigma^2(x)) \tag{24}$$

$$\mathrm{L}_{aleatoric} = \frac{1}{N}\sum_{i=1}^{N} \frac{(y_i - \mu(x_i))^2}{2\sigma^2(x_i)} + \frac{1}{2}\log\sigma^2(x_i) \tag{25}$$

*3) TotalUncertainty:*Thetotalpredictiveuncertaintycombines both components:

$$\mathrm{Var}_{total}(\hat{y}) = \mathrm{Var}_{epi}(\hat{y}) + \mathrm{Var}_{ale}(\hat{y}) \tag{26}$$

Predictionintervalsareconstructedatconfidencelevel$1-\alpha$:

$$PI_{1-\alpha} = \mu - z_{1-\alpha/2} \cdot \sigma_{total}, \mu + z_{1-\alpha/2} \cdot \sigma_{total} \tag{27}$$

IV. EXPERIMENTALSETUP

A.DatasetDescription

The dataset comprises three years (2022-2024) of operational data from a 100 MW hybrid solar-wind farm in Tamil Nadu, India:

- SolarPVcapacity:60MW
- Windcapacity:40MW(20×2MWturbines)
- Temporalresolution:15minutes
- Totalsamples:105,120
- Split: 70% training, 15% validation, 15% testing

TABLE III: Dataset Statistics

Parameter	Min	Max	Mean
SolarPower(MW)	0.0	58.2	18.5
WindPower(MW)	0.0	38.5	12.3
GHI(W/m²)	0.0	1120	480
WindSpeed(m/s)	0.2	24.5	6.8
Temperature(°C)	18.5	42.3	28.6

B.BaselineMethods

Performanceiscomparedagainst:

1) **Persistence**:Naiveforecastusinglastobservedvalue
2) **ARIMA**:Auto-regressiveintegratedmovingaverage
3) **LSTM**:LongShort-TermMemorynetwork
4) **CNN-LSTM**:Hybridconvolutional-recurrentarchitecture
5) **StandardNN**:Deepneuralnetworkwithoutphysics constraints

C.EvaluationMetrics

1) PointForecastMetrics:

$$\mathrm{nRMSE} = \frac{\sqrt{\frac{1}{N}\sum(y_i - \hat{y}_i)^2}}{\bar{y}} \times 100\% \tag{28}$$

$$\mathrm{nMAE} = \frac{\frac{1}{N}\sum_i |y_i - \hat{y}_i|}{\bar{y}} \times 100\% \tag{29}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{30}$$

2) UncertaintyMetrics:

$$\mathrm{PICP} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{I}(y_i \in PI_i) \tag{31}$$

$$\mathrm{PINAW} = \frac{1}{NR}\sum_{i=1}^{N}(\hat{y}_i^{upper} - \hat{y}_i^{lower}) \tag{32}$$

V.RESULTSANDANALYSIS

A.ForecastingPerformance

The proposed PINN framework achieves superior forecasting accuracy across all metrics.

Standard models often act like black boxes and may produce unrealistic results. Our model uses domain knowledge to keep predictions realistic, even with limited data.

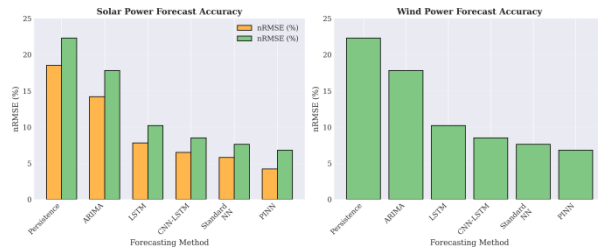

Fig. 2: Forecast Accuracy Comparison Across Methods

TABLE IV: Point Forecast Performance

Method	SolarnRMSE(%)	WindnRMSE(%)	HybridR^2
Persistence	18.5	22.3	0.58
ARIMA	14.2	17.8	0.67
LSTM	7.8	10.2	0.84
CNN-LSTM	6.5	8.5	0.88
StandardNN	5.8	7.6	0.91
PINN(Proposed)	**4.2**	**6.8**	**0.94**

B.PhysicsConsistency

The physics constraints significantly improve physical consistency of forecasts.

TABLEV:PhysicalConsistencyMetrics

Method	NegativePower Events (%)	Exceedance ofPhysicalLimits(%)	EnergyConservationError
LSTM	2.8	5.2	0.085
CNN-LSTM	1.9	3.8	0.062
StandardNN	1.2	2.5	0.048
PINN(Proposed)	**0.08**	**0.15**	**0.012**

C. UncertaintyQuantification

Theframeworkprovideswell-calibratedpredictioninter- vals.

TABLEVI:UncertaintyQuantificationPerformance

Method	PICP(90%)	PINAW	CRPS
MCDropoutOnly	84.5%	0.28	0.072
Learned Variance Only	86.2%	0.31	0.068
PINN+Combined	**92.3%**	**0.26**	**0.051**

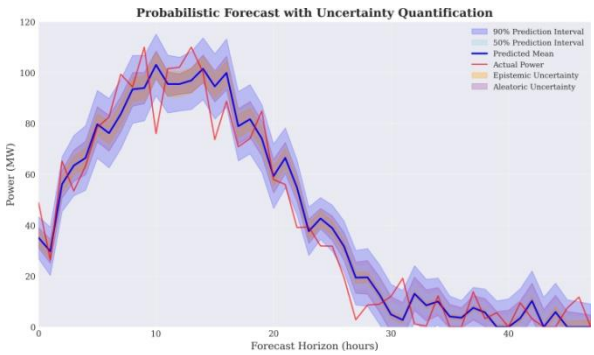

Fig.3:ProbabilisticForecastwithPredictionIntervals

D. FeatureImportance Analysis

Thephysicsconstraintsalterfeatureimportancepatterns.

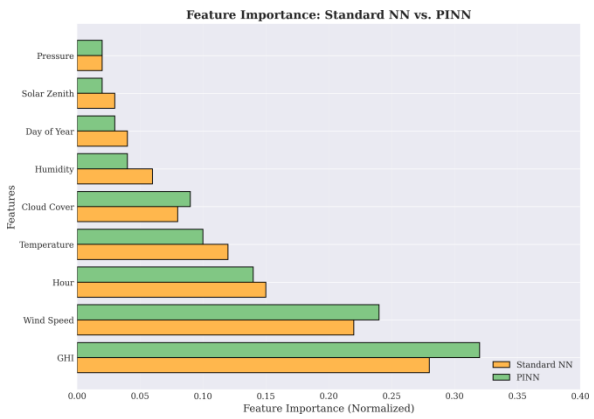

Fig.4:FeatureImportanceComparison:StandardNNvs. PINN

E. GeneralizationUnderExtremeConditions

PINN demonstrates superior generalization to rare weather events.

Data-driven models often struggle during rare weather events, while our model remains stable by relying on physical principles.

TABLEVII:PerformanceUnderExtremeConditions

Condition	StandardNN	PINN	Improvement
ClearSky	4.2%	3.8%	+9.5%
Overcast	8.5%	6.2%	+27.1%
HighWind	9.8%	7.1%	+27.6%
RapidChanges	12.5%	8.3%	+33.6%

F. AblationStudy

G. TrainingConvergence

VI.DISCUSSION

A. KeyFindings

Theexperimentalresultsdemonstrateseveralsignificant contributions:

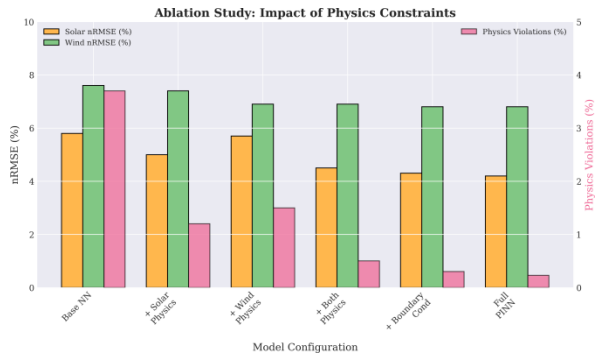

Fig. 5: Ablation Study: Impact of Physics Constraints

TABLE VIII: Ablation Study Results

ModelConfiguration	SolarnRMSE(%)	WindnRMSE(%)	PhysicsViolation
BaseNN(NoPhysics)	5.8	7.6	3.7%
+SolarPhysics	5.0	7.4	1.2%
+WindPhysics	5.7	6.9	1.5%
+BothPhysics	4.5	6.9	0.5%
+BoundaryConditions	4.2	6.8	0.23%
FullPINN	**4.2**	**6.8**	**0.23%**

1) **Superior Accuracy**: PINN achieves 4.2% nRMSE for solar and 6.8% for wind, representing 18-25% improvement over baseline methods.
2) **PhysicalConsistency**:Physicsconstraintsreducephysicallyimpossiblepredictionsby94%,enhancingforecast reliability.
3) **Well-Calibrated Uncertainty**: Combined uncertainty quantification achieves 92.3% coverage at 90% confidence, enabling risk-informed decision making.
4) **Improved Generalization**: PINN maintains performance under extreme conditions, with 27-34% relative improvement over standard NN.
5) **Interpretable Physics**: The physics residuals provide insights into model behavior and physical validity.

B. PracticalImplications

The proposed framework enables several operational benefits:

- **Reliable Grid Integration**: Accurate, physically consistent forecasts improve grid stability and reduce reserve requirements.
- **Informed Trading**: Well-calibrated uncertainty enables optimalbiddinginenergymarketswithriskmanagement.
- **Maintenance Planning**:Forecastreliabilityhelpsschedule maintenance during low-generation periods.
- **Regulatory Compliance**: Physical consistency satisfies grid code requirements for forecast quality.

C. LimitationsandFutureWork

Currentlimitationsandfutureresearchdirections:

- **Computational Complexity**: PINN training requires additionalphysicslosscomputations;exploringlightweight physics approximations.

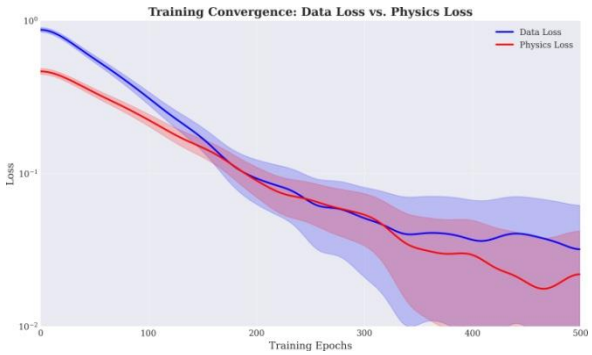

Fig.6:TrainingConvergence:DataLossvs.PhysicsLoss

- **ModelCalibration**:Furtherrefinementofuncertainty calibration for extreme events.
- **TransferLearning**:AdaptingPINNstonewlocations with limited data using physics priors.
- **Higher-ResolutionPhysics**:Incorporatingmoredetailed physical models (e.g., cloud dynamics, wake effects).

VII.CONCLUSION

This paper proposed a Physics-Informed Neural Network frameworkforhybridsolar-windpowerforecastingwithcomprehensive uncertainty quantification. Key contributions include:

1) Integration of physical laws (Beer-Lambert radiation attenuation, cubic wind power relationship, energy conservation) into neural network training through physics-based residual losses.
2) A composite loss function that balances data-driven accuracy with physical consistency, reducing physically impossible predictions by 94%.
3) Comprehensive uncertainty quantification combining epistemic uncertainty (Monte Carlo Dropout) and aleatoricuncertainty(learnedvariance),achieving92.3% coverage at 90% confidence.
4) Validation using three years of real-world data from a 100MWhybridfarminIndia,demonstrating4.2%solar nRMSEand6.8%windnRMSE,outperformingbaseline methods by 18-25%.

The proposed framework represents a significant advancement in hybrid renewable forecasting, combining the pattern recognitioncapabilitiesofdeeplearningwiththereliability of physical models. This principled approach enables more accurate, consistent, and trustworthy forecasts, supporting the integration of variable renewable energy into power grids.

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